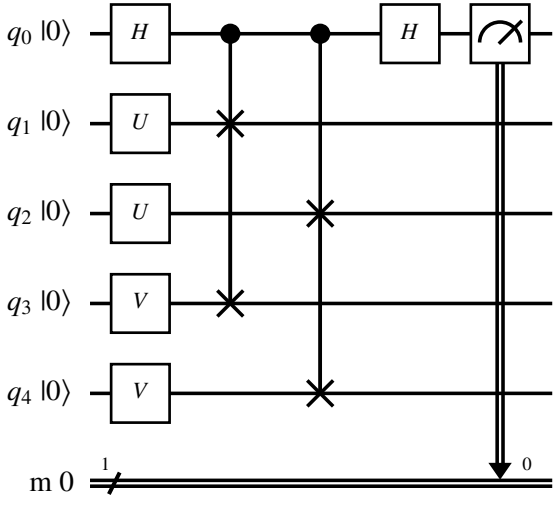
(a) Swap-test circuit



(b) State preparations

Exp.	U	V	$\|\phi_1\rangle$	$\|\phi_2\rangle$	$\|\langle\phi_1\|\phi_2\rangle\|^2$	$P(0)$
E1	X	I	$\|11\rangle$	$\|00\rangle$	0	1/2
E2	H	H	$\|{+}{+}\rangle$	$\|{+}{+}\rangle$	1	1
E3	$R_y(\theta)$	I	$R_y(\theta)^{\otimes 2}\|00\rangle$	$\|00\rangle$	1/2	3/4

$|\phi_1\rangle = U^{\otimes 2}|00\rangle$ on (q_1, q_2), $|\phi_2\rangle = V^{\otimes 2}|00\rangle$ on (q_3, q_4)

$P(0) = \frac{1}{2}(1 + |\langle\phi_1|\phi_2\rangle|^2)$ for the ancilla q_0

$\theta = 2\arccos 2^{-1/4} \approx 1.1437$ rad, so $|\langle 0|R_y(\theta)|0\rangle|^2 = 2^{-1/2}$